University of St Andrews | FOUNDED 1413

Higher Education Achievement Report (HEAR)

incorporating the Diploma Supplement

for Undergraduates

Barnabas Nandor Tanczos

This Higher Education Achievement Report (HEAR) follows the model developed by the European Commission, Council of Europe and UNESCO/CEPES for the Diploma Supplement. The purpose of the Diploma Supplement is to provide sufficient recognition of qualifications (diplomas, degrees, certificates, etc.). It is designed to provide a description of the nature, level, context and status of the studies that were pursued and successfully completed by the individual named on the original qualifications. It should be free from any value judgements, equivalence statements or suggestions about recognition.

Glossary of terms

The following abbreviations, acronyms and terms are used in this HEAR.

ECTS (European Credit Transfer and Accumulation System)
The system of module credit transfer which helps make teaching and learning in Higher Education more transparent across Europe and facilitates the recognition of all studies. The University of St Andrews adopted a modularised scheme for all degree programmes in 1994. All modules at the University of St Andrews have an accreditation in the ECTS scheme. Since St Andrews operates on an annual load of 120 credits and ECTS one of 60 credits, credits transferred to the other institution will be exactly half the credits shown for each module. See www.st-andrews.ac.uk/media/honintroappendix-10-11.pdf

HESA (Higher Education Statistics Agency)
The official agency for the collection, analysis and dissemination of quantitative information about higher education in the United Kingdom. Students are assigned a unique national identifying number when registered at institutions of Higher Education in the UK. Further details can be found at www.hesa.ac.uk

SCQF (Scottish Credit and Qualification Framework)
The national credit transfer system for all levels of qualifications in Scotland. Further details can be found at www.scqf.org.uk

St Andrews GPA (Grade Point Average)
The St Andrews GPA for undergraduates is the credit-weighted mean of Honours programme grades.

The St Andrews GPA is calculated during classification and is displayed at the same time as the degree classification and/or degree award.

The St Andrews GPA uses a 20-point scale in common with classification and module final assessments at the University.

The scale with descriptors is available at: www.st-andrews.ac.uk/staff/teaching/examinations/honours

Further Information on the University of St Andrews

Structure and governance
www.st-andrews.ac.uk

Reporting scale and grading system
www.st-andrews.ac.uk/administration/academicdatateam/assessmentandawards/gradingsheets/

Senate regulations for the awarding of degrees
www.st-andrews.ac.uk/students/rules/ugsenateregulations/

This Higher Education Achievement Report is printed in black ink on paper watermarked with the seal of the University in grey and carries the official University stamp. It is not valid unless in this format.

1 Holder of qualification

Family name(s)	**Tanczos**
Given name(s)	**Barnabas Nandor**
Date of birth	**29 April 2003**
Student identification number	**210018010**
HESA identification number	**2111730009501**

2 Qualification achieved

Qualification achieved	**Bachelor of Science (Honours)**
Classification achieved	**First Class**
Main field(s) of study	**Computer Science and Mathematics**
Institution(s) administering studies	**University of St Andrews**
Language(s) of instruction/assessment	**English**
Awarding institution(s)	**University of St Andrews**

3 Qualification levels

HESA level of qualification	**First degree with Honours**
National qualifications framework level	**SCQF 10**
Official length of programme	**4 years if mode of study is Full Time**
Programme entry requirements or access	**www.st-andrews.ac.uk/prospectus**

4 Qualification - Mode of study, programme requirements and results gained

4a Mode and location of study

2021/2	**Full-time - Studying at University of St Andrews**
2022/3	**Full-time - Studying at University of St Andrews**
2023/4	**Full-time - Studying at University of St Andrews**
2024/5	**Full-time - Studying at University of St Andrews**

4b General programme requirements

https://www.st-andrews.ac.uk/subjects/reqs

4c Specific programme requirements at 3000 level and above

https://www.st-andrews.ac.uk/subjects/reqs

4d Programme specifications

https://www.st-andrews.ac.uk/subjects/specs/

4e Modules taken and results gained by Barnabas Nandor Tanczos (210018010/1)

Key to grades: A = Audited, D = Deferred, E = Exempt; F = Failed, M = Decanal intervention, P = Passed, PC = Grade capped, S = Special circumstances, SP = Special circumstances due to COVID-19 pandemic, V = Void - mitigating circumstances, X = Failed requirements, XC = Failed - right to retake module for credit only, XN = Failed - no right to take additional module, XR = Failed - right to take another module for credit only, Z = Unreported

Year	Programme and main fields of study / Module level, code and title		University of St Andrews credits possible	credits counted	ECTS credits gained	Module Assessments Grade / Result Attempt 1	Attempt 2
2021/2	Bachelor of Science (Honours) Computer Science and Mathematics						
1000	CS1002	Object-Oriented Programming	20	20	10	20.0 P	
1000	CS1003	Programming with Data	20	20	10	19.1 P	
1000	CS1006	Programming Projects	20	20	10	18.0 P	
1000	MT1002	Mathematics	20	20	10	20.0 P	
1000	MT1003	Pure and Applied Mathematics	20	20	10	17.0 P	
1000	PY1011	Moral and Political Controversies	20	20	10	17.3 P	
	Credits for academic session		**120**	**120**	**60**		
2022/3	Bachelor of Science (Honours) Computer Science and Mathematics						
2000	CS2001	Foundations of Computation	30	30	15	18.4 P	
2000	CS2002	Computer Systems	30	30	15	18.3 P	
2000	MT2501	Linear Mathematics	15	15	7.5	19.0 P	
2000	MT2502	Analysis	15	15	7.5	19.0 P	
2000	MT2504	Combinatorics and Probability	15	15	7.5	18.0 P	
2000	MT2508	Statistical Inference	15	15	7.5	17.0 P	
	Credits for academic session		**120**	**120**	**60**		
2023/4	Bachelor of Science (Honours) Computer Science and Mathematics						
3000	CS3050	Logic and Reasoning	15	15	7.5	19.2 P	
3000	CS3052	Computational Complexity	15	15	7.5	19.7 P	
3000	CS3099	Software Engineering Team Project	30	30	15	17.1 P	
3000	MT3502	Real Analysis	15	15	7.5	17.8 P	
3000	MT3507	Mathematical Statistics	15	15	7.5	19.0 P	
3000	MT3508	Applied Statistics	15	15	7.5	17.8 P	
4000	MT4004	Real and Abstract Analysis	15	15	7.5	19.6 P	
	Credits for academic session		**120**	**120**	**60**		
2024/5	Bachelor of Science (Honours) Computer Science and Mathematics						
3000	CS3105	Artificial Intelligence	15	15	7.5	19.2 P	
3000	MT3503	Complex Analysis	15	15	7.5	19.6 P	
4000	CS4098	Minor Software Project	15	15	7.5	19.0 P	
4000	CS4303	Video Games	15	15	7.5	18.8 P	
4000	MT4515	Functional Analysis	15	15	7.5	16.3 P	
4000	MT4531	Bayesian Inference	15	15	7.5	16.3 P	
4000	MT4599	Project in Mathematics / Statistics	15	15	7.5	17.5 P	
5000	CS5020	Principles of Computer Communication Systems	15	15	7.5	18.0 P	
	Credits for academic session		**120**	**120**	**60**		
	Total credits counted on this programme		**480**	**480**	**240**		
	St Andrews GPA (Grade Point Average)		**18.3 out of 20.0 possible**				

Final award of **Bachelor of Science (Honours) Computer Science and Mathematics**

First Class

Awarded on **03 July 2025**

5 Qualification - Function and further access

This qualification may allow access to further study (at SCQF 11 or for equivalent EHEA second cycle qualifications) subject to the individual requirements of the institution concerned.

The following professional bodies or institutions have accredited this qualification:

None known or applicable.

6 Additional achievements

Students at the University of St Andrews have the opportunity to engage in both co-curricular and extra-curricular activities which may contribute to the life of the University and the wider community as well as to their own personal and professional development. The achievements reported here have been verified by the University of St Andrews and information on the approvals process used to verify data for inclusion in this section can be found at www.st-andrews.ac.uk/students/awards

Academic Prizes

2024/5 Deans' List
Awarded to students who average above 16.5 in all modules across the academic year

2023/4 Deans' List
Awarded to students who average above 16.5 in all modules across the academic year

2023/4 Medal (Statistics BSc Third level)
Awarded for performance in Third level Statistics (BSc)

2022/3 Deans' List
Awarded to students who average above 16.5 in all modules across the academic year

2021/2 Deans' List
Awarded to students who average above 16.5 in all modules across the academic year

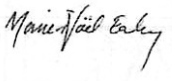
2021/2 Medal (Module CS1002 Computer Science)
Awarded for performance in Module CS1002 - Computer Science

2021/2 Medal (Module CS1006 Programming Projects)
Awarded for performance in Module CS1006 - Programming Projects

Scholarships / Grants / Bursaries

2021/2 Global Merit Award
In recognition of the quality and academic excellence of my application to the programme

7 Certification

Name of signatory	**Marie-Noël Earley**			Registry
Official capacity	**Academic Registrar**	Signature		
Date certified	**18th of June 2025**			

Followed by:

A national descriptor of Scottish higher education with a summary of SCQF levels, "Description of Higher Education in Scotland" endorsed by a national committee, The Quality Working Group.

Description of Higher Education in Scotland

Introduction

Scotland has a distinctive higher education system and also operates under a devolved government, which includes devolved responsibility for higher education. There is a separate Description of Higher Education in England, Wales and Northern Ireland where the system is different to that of Scotland.

Scotland's distinctive higher education system has 19 higher education institutions (HEIs). The 15 universities, the Open University in Scotland, a college of higher education, an art school, and a conservatoire all receive funding for research and for learning and teaching through the Scottish Funding Council (see www.sfc.ac.uk); funding is also received from other sources.

The HEIs are independent, self-governing bodies, active in teaching, research and scholarship. Where HEIs are degree awarding bodies they design the curriculum for the degrees they award, set the conditions on which they are awarded and the admissions arrangements. Degrees and other higher education qualifications are legally owned by the awarding institution, not by the state.

The HEIs offer qualifications at undergraduate (Bologna first cycle) and postgraduate (Bologna second and third cycle) levels. In Scotland, the law distinguishes the power to award degrees on the basis of completion of taught programmes (bachelors and most masters degrees) from the power to award Research (doctoral) Degrees. Most universities have powers to award taught and research Degrees. Some other HEIs have powers to award taught Degrees while others offer programmes leading to Degrees awarded by HEIs with Degree awarding powers.

Lists of institutions with powers to award degrees and institutions recognised by authorities in Scotland as being able to offer courses leading to a Degree of another HEI may be found at www.universities-scotland.ac.uk.

A small number of taught Degrees are available in tertiary colleges by the authority of a duly empowered HEI.

Qualifications

The types of qualifications awarded at undergraduate (first cycle) and postgraduate level (second and third cycles) in Scotland are described in "The Framework for qualifications of higher education institutions in Scotland" which includes qualifications descriptors, developed with the university sector and published by the Quality Assurance Agency (QAA) (www.qaa.ac.uk). The Framework was self-certified as compatible with the Framework for Qualifications of the European Higher Education Area, the qualifications framework adopted as part of the Bologna Process, in October 2006. The Framework is also an integral part of a wider national framework: the Scottish Credit and Qualifications Framework (SCQF) that covers all forms of programmes and qualifications from school to to doctorates (see Table 1 and www.scqf.org.uk).

Institutions use SCQF levels and credit points for students entering or transferring between programmes or institutions, and use ECTS for transfers within the European area.

Admission

Admissions requirements for particular programmes are set by the HEIs which offer a range of routes for entry and/or credit transfer into their programmes, and admit students whom they believe have the potential to complete their programmes successfully. The Open University is an open entry institution.

The most common qualification for entry to higher education is the Higher and, for a small number of high tariff courses, the Advanced Higher or, for entrants from the rest of the UK, the General Certificate of Education at "Advanced" level or comparable qualifications. Pupils seeking to enter a HEI would normally take a number of Highers at an appropriate stage in the Senior Phase (4th, 5th and 6th year) of secondary school, according to the requirements of their own learning journey, or at a tertiary college. HEIs usually require 4-6 Highers for entry, but this may vary with subject. Highers are studied in considerable depth, involving coursework and final examinations. Advanced Highers have historically been taken by some pupils in S6 as a means of extending their specialisation, normally following successful completion of a Higher in that subject. They are also available in some tertiary colleges. Pupils may also take a Scottish Baccalaureate in Sciences, Languages, Expressive Arts, or Social Sciences and these consist of related Highers and Advanced Highers and an interdisciplinary project. Another major route into Degrees, often with full transfer of credit, is from Higher National Qualifications offered in tertiary colleges.

Quality Assurance

The academic standards of qualifications are secured and the quality of the student learning experience enhanced by the HEIs using a range of processes including extensive use of external examiners and suitably qualified independent external individuals. In some subject areas, Professional, Statutory and Regulatory Bodies (PSRBs) have a role to ensure that programmes meet the needs and standards of a particular profession; PSRBs do not set or regulate the academic standards of awards, which is the responsibility of the Degree awarding body.

HEIs in Scotland demonstrate their public accountability for quality and standards through a national Quality Enhancement Framework which assures academic standards and the quality of learning experiences. It has a strong focus on enhancement as follows:

HEIs take account of the UK Quality Code for Higher Education, which is published by the QAA. The Quality Code is a UK-wide code of practice for quality assurance and enhancement, which includes qualifications frameworks and UK subject level "benchmark statements" as well as extensive guidance on the quality of the student learning experience and provision of public information (see www.qaa.ac.uk). Higher Education providers use the Quality Code to design their respective policies for maintaining academic standards and to enhance quality. Reviewers use it as a key reference point for the external review and quality assurance of HEIs.

Subject level quality reviews are conducted by HEIs in accordance with guidance issued by the Scottish Funding Council (see www.sfc.ac.uk) and in light of the Quality Code.

External reviews of HEIs are conducted by the Quality Assurance Agency for Higher Education in Scotland (QAA Scotland). QAA Scotland is an independent body and charity established to provide public confidence in the quality and standards of higher education. The method of external review in Scotland involves teams of peer reviewers, including student and international reviewers. QAA Scotland publishes reports on the outcome of reviews and makes judgements about the effectiveness of the HEIs' arrangements for assuring academic standards and enhancing the quality of the learning experiences offered (see www.qaa.ac.uk). QAA Scotland also manages a programme of national Enhancement Themes (see www.enhancementthemes.ac.uk).

A national development service supports students in their role as active participants in assuring standards and enhancing quality (see www.sparqs.org.uk).

[1] Awarded by the Scottish Qualifications Authority

TABLE 1

The Scottish Credit and Qualifications Framework (SCQF)

The SCQF covers all the major qualifications in Scotland from school to Doctorate and including work-based Scottish Vocational Qualifications (SVQs)

SCQF Level	Qualifications of Higher Education Institutions	SQA Higher National and National Units, Courses and Group Awards	SVQs
12	Doctoral Degrees (Minimum 540 SCQF credits)		Professional Apprenticeship
11	Masters Degrees (Minimum 180 SCQF credits), Integrated Masters Degrees (Minimum 600 SCQF credits, Postgraduate Diploma (Minimum 120 SCQF credits), Postgraduate Degree with Honours (Minimum 60 SCQF credits)		Professional Apprenticeship SVQ 5
10	Bachelors Degree with Honours (Minimum 480 SCQF credits) Graduate Diplomas and Certificates		Professional Apprenticeship
9	Bachelors Degree (Minimum 360 SCQF credits) Graduate Diplomas and Certificates	Professional Development Award	Technical Apprenticeship SVQ 4
8	Diploma of Higher Education (Minimum 240 SCQF credits)	Higher National Diploma	Technical Apprenticeship SVQ 4
7	Certificate of Higher Education (Minimum 120 SCQF credits)	Advanced Higher,Scottish Baccalaureate, Higher National Certificate	
6		Higher	Modern Apprenticeship SVQ 3
5		National 5 Intermediate 2	Modern Apprenticeship SVQ 2
4		National 4 Intermediate 1	SVQ 1
3		National 3 Access 3	
2		National 1 Access 2	
1		National 1 Access 1	

Notes

1 SCQF levels represent increasing complexity and demand in learning outcome.

2 One credit represents the outcomes achievable by the average student though 10 notional hours of learner effort. In general terms, one full-time undergraduate year is considered to be 120 credits worth of learning. A postgraduate year is 180 credits. 1 ECTS credit is deemed equivalent to 2 SCQF credits. Research degrees - Master of Philosophy (MPhil) and Doctor of Philosophy (PhD) are not credit rated.

3 Graduate Certificates (minimum of 60 SCQF credits) and Graduate Diplomas (minimum of 120 credits) are offered at levels 9 and 10 within the SCQF framework. They are offered for programmes that are for graduates but do not have outcomes that are at postgraduate level.

4 The Bachelors Degree (level 9) leads to employment and in some instances can give access to postgraduate study particularly when accompanied by relevant work or professional experience.

5 At postgraduate levels, the framework and the higher education qualifications are the same as those for the rest of the UK. The Honours Degree levels of the two frameworks are considered to be in broad alignment (the Honours Degree in Scotland normally takes 4 years and that in the rest of the UK takes 3 years). Below Honours level the frameworks reflect the different educational structures of Scotland and the rest of the UK).

This national description is endorsed by the Quality Working Group which is a national committee with members from The Quality Assurance Agency for Higher Education, Scotland; The Scottish Funding Council; Universities Scotland and the National Union of Students in Scotland.